Supplementary Materials

A compact workflow for determining umbrella species

(Hartono et al.)

Appendix S1 — Literature search and screening protocol

The initial species pool was derived from the Baluran National Park camera-trap records. For body-size screening, adult mammals weighing > 5 kg were classified as large-bodied mammals, whereas those < 5 kg were classified as small-bodied mammals, following Anderson and Jones (1967, as cited in Maharadatunkamsi et al., 2020). Species body-mass values were compiled from species-specific published references. Candidate umbrella species were then retained from the camera-trap records if they were classified as large-bodied and had ≥ 5 independent detections.

We searched Scopus, Google Scholar, GARUDA, and Neliti between December 2024 and January 2025 for peer-reviewed information on the candidate mammal species recorded by camera-traps in Baluran National Park. We also consulted the IUCN Red List and Baluran National Park management documents for species-status and management-context information. Searches included studies published up to 2024 in English or Bahasa Indonesia.

Searches combined (i) the scientific name and common or vernacular synonyms of each species, (ii) topic terms, and (iii) geographic terms (Java or Indonesia). Quotation marks were used for multi-word phrases to retrieve exact matches (e.g., "Javan leopard", "Macan tutul jawa"), whereas single-word terms were searched without quotation marks to capture spelling variants and broader usage (e.g., banteng). The following base query structure was used: [scientific OR common/vernacular synonyms] AND [topic terms] AND [Java OR Indonesia].
Examples:

1. (*Bos javanicus javanicus* OR banteng) AND (biology OR ecology OR home range OR wilayah jelajah OR range size OR movement) AND (Java OR Indonesia).
2. (*Panthera pardus melas* OR "Javan leopard" OR "Macan tutul jawa") AND (biology OR ecology OR disturbance OR gangguan OR human activity OR aktivitas manusia) AND (Java OR Indonesia).
   Synonyms for other species included: dhole/ajag (*Cuon alpinus javanicus*), Javan rusa/rusa timor (*Rusa timorensis russa*), muntjac/kijang muncak (*Muntiacus muntjak muntjak*), wild boar/babi hutan (*Sus scrofa*), long-tailed macaque/monyet ekor panjang (*Macaca fascicularis fascicularis*), and Javan langur/ebony leaf monkey/lutung budeng (*Trachypithecus auratus*).

"Relevant to Baluran" refers to studies conducted in Baluran National Park or explicitly reporting Baluran populations. "Relevant to Java" refers to studies conducted in Java or explicitly reporting Javan populations or subspecies. When Baluran- and Java-specific evidence was unavailable, we used studies from the Sundaic region or from seasonally dry tropical forest, savanna, or savanna–forest mosaic systems supporting terrestrial mammal communities under pressures relevant to Baluran, namely grazing, resource extraction, hunting/poaching, or dry-season habitat degradation.

We included peer-reviewed empirical studies and reviews, prioritising evidence from Baluran National Park, then from Java or studies explicitly referring to Javan populations or subspecies. When such evidence was unavailable, we used secondary evidence sources that met the operational

criteria above. We excluded captive studies, veterinary-only reports, and livestock or production-oriented studies unless they informed free-ranging ecology relevant to Baluran. Duplicates across databases and search strings were removed prior to extraction.

We screened titles and abstracts and then assessed full texts for eligibility. The numbers of records retrieved, screened, and included after deduplication are reported in Appendix Table S1. Extracted values and notes were logged in a spreadsheet. Indicator assignment rules and scoring rubrics are reported in Appendix S2/Table S2, and the evidence table supporting species-level scoring is provided in Appendix S4.

Table S1. Publications consulted for indicator scoring (C2, C3, C6), grouped by species. *Citations listed as Author et al., year; DOI/URL. DOI URLs were shortened to DOI strings (e.g., https://doi.org/10.xxxx -> 10.xxxx). All searches and source compilation were conducted during December 2024–January 2025.*

Species	C2 (Knowledge Base)	C3 (Home Range)	C6 (Disturbance Sensitivity)
Bos javanicus javanicus	Gardner et al., 2014 (10.1017/CBO9781139568098.015); Muhammad et al., 2007 (10.22146/jik.1553); Pudyatmoko et al., 2006 (10.3923/jbs.2006.370.374); Pudyatmoko et al., 2007 (10.3923/jbs.2007.854.859); Rahman et al., 2019 (10.4308/hjb.26.2.68); Rahman, 2020 (10.12933/therya-20-840); Santosa & Delfiandi, 2007 (https://media.neliti.com/media/publications/231525-analisis-pola-penggunaan-ruang-dan-wilay-f66cd505.pdf); Sawitri et al., 2023 (10.55981/brin.602.c620); Siddiq et al., 2022 (10.2991/978-94-6463-062-6_6); Qiptiyah et al., 2019 (10.2991/978-94-6463-062-6_6); Qiptiyah et al., 2023 (10.55981/brin.602.c619).	Santosa & Delfiandi, 2007 (https://media.neliti.com/media/publications/231525-analisis-pola-penggunaan-ruang-dan-wilay-f66cd505.pdf)	Pudyatmoko et al., 2017 (10.1016/j.mambio.2017.04.001)
Bubalus bubalis	Giorgio et al., 2017 (10.2174/9781681084176117010101); Heinen, 1996 (10.1046/j.1523-1739.1996.10040931.x); Sutomo et al., 2015 (10.13140/RG.2.1.2552.4883).	Napolitano et al., 2017 (10.2174/9781681084176117010101)	Pudyatmoko et al.,2017 (10.1016/j.mambio.2017.04.001)
Cuon alpinus javanicus	Cohen, 1978 (10.2307/3503800); Havmøller et al., 2022 (10.1002/ece3.11666); Iyengar et al., 2005 (10.1111/j.1365-294X.2005.02582.x); Nurvianto et al., 2015 (https://bepls.com/nov_2015/7BEPLS%20(1).pdf); Nurvianto et al., 2015 (https://www.biotaxa.org/ajcb/issue/view/8653); Nurvianto et al., 2016 (10.47605/tapro.v8i1.210); Nurvianto et al., 2021 (10.2991/absr.k.220406.088); Rahman et al., 2018 (10.25225/fozo.v67.i3-4.a8.2018); Sillero et al., 2004 (https://portals.iucn.org/library/sites/library/files/documents/1990-008.pdf).	Nurvianto et al., 2015 (https://bepls.com/nov_2015/7BEPLS%20(1).pdf)	Nurvianto et al., 2015 (https://bepls.com/nov_2015/7BEPLS%20(1).pdf)
Macaca fascicularis fascicularis	Fuentes et al., 2005 (10.1002/ajp.20138); Gumert, 2010 (10.1007/978-1-4419-1560-3_18); Hansen et al., 2015 (https://www.researchgate.net/publication/331634373_The_interactions_between_long-tailed_macaques_Macaca_fascicularis_and_tourists_in_Baluran_National_Park_Indonesia); Hansen et al., 2019 (10.1111/csp2.88); Hansen et al., 2020 (10.1163/18759866-bja10006); Hasan et al., 2023 (10.1007/s10329-023-01062-z); Setiawan et al., 2024 (10.35438/conserva.v2i1.204).	Hansen et al., 2015 (https://www.researchgate.net/publication/331634373_The_interactions_between_long-tailed_macaques_Macaca_fascicularis_and_tourists_in_Baluran_National_Park	Hansen et al., 2015 (https://www.researchgate.net/publication/331634373_The_interactions_between_long-tailed_macaques_Macaca_fascicularis_and_tourists_in_Baluran_National_P

Species	C2 (Knowledge Base)	C3 (Home Range)	C6 (Disturbance Sensitivity)
		_Indonesia)	ark_Indonesia)
Muntiacus muntjak	Ekanasty et al., 2014 (10.29243/medkon.19.3.%p); Farida et al., 2003 (10.13057/biodiv/d040206); Farida et al., 2006 (10.3759/tropics.15.371); Pudjirahaju et al., 2015 (https://ojs.unud.ac.id/index.php/jvet/article/download/13324/9011); Rahmadia et al., 2023 (https://jurnal.uny.ac.id/index.php/jsd/article/download/59156/pdf/185552); Rahman et al., 2017 (10.1016/j.actao.2017.05.008).	McCullough et al., 2000 (10.2307/3803241); Rahman et al., 2017 (10.1016/j.actao.2017.05.008).	Rahman et al., 2017 (10.1016/j.actao.2017.05.008)
Panthera pardus melas	Ardiansyah et al., 2020 (10.20886/jphka.2020.17.2.127-142); Ario et al., 2018 (https://pdfs.semanticscholar.org/7208/aobaaob2e49a754b62c55008a4df9176d342.pdf); Ario et al., 2022 (10.4081/tz.2022.120); Ario et al., 2022 (10.11609/jott.7483.14.7.21331-21346); Ariyanto et al., 2024 (10.1016/j.gecco.2024.e03020); Gippoliti et al., 2007 (10.1163/18759866-07601005); Gunawan et al., 2012 (10.13140/RG.2.2.33183.97444); Gunawan et al., 2013 (https://www.academia.edu/37519677/Bio_Ekologi_dan_Konservasi_KARNIVORA_Spesies_Kunci_yang_Terancam_Punah_Bio_Ekologi_dan_Konservasi_KARNIVORA_Spesies_Kunci_yang_Terancam_Punah); Gunawan et al., 2017 (10.20886/jphka.2017.14.1.35-44); Gunawan et al., 2020 (10.13057/biodiv/d180229); Gunawan et al., 2022 (10.29244/medkon.27.3.128-139); Kholiq et al., 2024 (10.19184/lsb.v2i1.47666); Pudyatmoko, 2019 (10.1007/s13364-018-0391-z); Rahman et al., 2018 (10.25225/fozo.v67.i3-4.a8.2018); Wibisono et al., 2018 (10.1371/journalpone.0198369).	Wibisono et al., 2018 (10.1371/journalpone.0198369)	Pudyatmoko et al., 2017 (10.1016/j.mambio.2017.04.001)
Rusa timorensis russa	Fawzy et al., 2019 (10.1088/1755-1315/394/1/012015); Latifah et al., 2022 (10.24252/bio.v10i2.30712); Rahman et al., 2017 (10.1016/j.actao.2017.05.008); Rahman et al., 2020 (10.1016/j.jnc.2020.125795); Yuliawati et al., 2018 (10.1088/1757-899X/434/1/012107).	Rahman et al., 2020 (10.1016/j.jnc.2020.125795)	Rahman et al., 2017 (10.1016/j.actao.2017.05.008)
Sus scrofa	Albert et al., 2014 (10.25077/jbioua.3.3.%25p.2014); Suripto, 2020 (https://core.ac.uk/download/pdf/298729972.pdf).	Pinsky et al., 2019 (10.1575/1912/bco-dmo.752795.1)	Garza et al., 2018 (10.1093/jmammal/gyx154)

Species	C2 (Knowledge Base)	C3 (Home Range)	C6 (Disturbance Sensitivity)
Trachypithecus auratus	Al-Huda et al., 2024 (10.13057/biodiv/d250419); Eliana et al., 2017 (10.20884/1.SB.2017.4.2.403); Hendrawan et al., 2019 (10.13057/psnmbi/m050243); Leca et al., 2013 (10.1896/052.026.0106); Maulahila et al., 2023 (10.2991/978-94-6463-062-6_15); Nijman, 2000 (10.1163/18759866-06903002); Rahman et al., 2021 (10.1088/1755-1315/948/1/012006); Tsuji et al., 2013 (10.1007/s10329-012-0334-2); Tsuji et al., 2019 (10.1016/j.mambio.2019.08.001); Wardhana et al., 2022 (10.13057/biodiv/d230845).	Leca et al., 2013 (10.1896/052.026.0106)	Rahman et al., 2021 (10.1088/1755-1315/948/1/012006)

Appendix S2 — Indicator definitions and scoring rules (C1–C7)

Table S2. Seven indicators were used with equal weights and explicit mappings. C1 tertile thresholds for this dataset: Q1 = 3.1994 %, Q2 = 4.9107 % (with ties handled conservatively: a value exactly equal to Q2 is assigned Score 2).

Code	Criterion	Operational Definition & Data Source	Scoring Rule (1–3)
C1	Species co-occurrence	Site-level co-occurrence across camera stations (park-wide, N = 56), summarized as mean *Oij/N* across other mammals; companion hypergeometric check (R `cooccur`).	3: > 4.9107 % | 2: > 3.1994–≤ 4.9107 % | 1: ≤ 3.1994 %
C2	Natural history & ecology well-known	Number of peer-reviewed studies relevant to Baluran National Park, Java, or explicitly defined analogous systems.	3: ≥ 10 | 2: 6–9 | 1: ≤ 5
C3	Home-range size	Largest reported home range (km²) for each species, prioritizing estimates from Java or analogous systems.	3: > 5.56 | 2: > 2.44–≤ 5.56 | 1: ≤ 2.44
C4	Persistence (IUCN risk proxy)	Uses IUCN Red List category as a standardized proxy for relative persistence (inverse of extinction risk) in IUCN Red List Categories and Criteria v3.1.	3: LC | 2: NT | 1: VU/EN/CR/NE
C5	Management needs that benefit co-occurring species	Management relevance was assigned from Baluran management documents and implementation reports. We reviewed the Baluran Long-term Management Plan 2024–2033, which targets +20% banteng and +10% Javan leopard by 2033 and increased banteng presence in the savanna. Scores followed an explicit rubric (Ardiantiono et al., 2024): 3 = management priority; 2 = management concern; 1 = limited management attention.	3: Priority species | 2: Management concern | 1: Limited concern
C6	Sensitivity to human disturbance	Based on published and local evidence of avoidance/tolerance responses to human pressure (roads/settlements, grazing, extraction, hunting), including changes in detection/occupancy with disturbance covariates where available.	3: Moderate | 2: High | 1: Low
C7	Sampling & monitoring feasibility	Feasibility reflects whether a species can be reliably detected at the park's routine survey effort using standard methods (camera-traps, sign surveys, or direct observation): 3 = consistently detectable with cameras and/or diagnostic signs; 2 = detectable but typically requires targeted placement/effort; 1 = rarely detected or detection is unreliable under routine monitoring, based on our camera-trap detections and supporting literature/park records.	3: Easy | 2: Moderate | 1: Difficult

Appendix S3 — Camera-trap detections and presence–absence

Table S3. Independent detection events by species (events; ≥ 30-min separation).

Species	*Total independent detections*
Bos javanicus javanicus	59
Bubalus bubalis	19
Cuon alpinus javanicus	12
Macaca fascicularis fascicularis	33
Muntiacus muntjak muntjak	28
Panthera pardus melas	13
Rusa timorensis russa	97
Sus scrofa	35
Trachypithecus auratus	13

Note:

- Counts are events, not number of stations.
- Presence–absence by station is provided as a CSV (co_occur_bnp_56.csv), available in the data repository: https://github.com/drajat313/baluran-umbrella-spp-data

Appendix S4 — Composite seven-indicator scores and tiers (C1–C7) for nine mammal candidates in Baluran National Park

Table S4. Per-species scores (1–3) for seven-indicators, unweighted totals, and priority tier (≤ 33rd = Low; 34th–66th = Moderate; ≥ 67th = High). (Ardiantiono et al., 2024; Caro, 2010; Fleishman et al., 2000; Seddon and Leech, 2008; Thompson et al., 2021; Wilson et al., 2010)

No	***Species***	***Total Score***	***Co-Occurrence (%) (C1)***	***Score***	***Bio Ecological Knowledge (C2)***	***Score***	***Home Range (C3)***	***Score***	***IUCN Status (C4)***	***Score***	***Management Priority (C5)***	***Score***	***Sensitivity to Human Disturbance (C6)***	***Score***	***Ease of Monitoring (C7)***	***Score***
1	*B. javanicus javanicus*	**16**	4.91%	2	Well studied species (10)	3	3.37 km² (Santosa and Delfiandi, 2007)	Intermediate (2)	CR	1	Priority species	3	High	2	Easy	3
2	*P. pardus melas*	**16**	0.45%	1	Well studied species (15)	3	13.6 km² (Wibisono et al., 2018)	Large (3)	EN	1	Priority species	3	High	2	Easy	3
3	*C. alpinus javanicus*	**15**	2.01%	1	Moderately studied species (9)	2	14.18 km² (Nurvianto et al., 2015)	Large (3)	EN	1	Management concern	2	Moderate	3	Easy	3
4	*M. muntjak muntjak*	**15**	4.92%	3	Moderately studied species (6)	2	1.68 km² (McCullough et al., 2000)	Small (1)	LC	3	Management concern	2	High	2	Moderate	2
5	*R. timorensis russa*	**15**	7.14%	3	Less studied species (5)	1	5 km² (Rahman et al., 2020)	Intermediate (2)	VU	1	Management concern	2	Moderate	3	Easy	3

6	*S. scrofa*	**15**	3.57%	2	Less studied species (2)	1	7 km² (Pinsky and McCauley, 2019)	Large (3)	LC	3	Management concern	2	Low	1	Easy	3
7	*B. bubalis*	**12**	3.57%	2	Less studied species (3)	1	2.83 km² (Napolitano et al., 2017)	Intermediate (2)	NE	1	Limited management concern	1	High	2	Easy	3
8	*M. fascicularis fascicularis*	**12**	5.80%	3	Moderately studied species (7)	2	1.75 km² (Hansen et al., 2015)	Small (1)	EN	1	Limited management concern	1	Low	1	Easy	3
9	*T. auratus*	**12**	2.46%	1	Well studied species (10)	3	0.14 km² (Leca et al., 2013)	Small (1)	VU	1	Management concern	2	High	2	Moderate	2

Notes: *Bubalus bubalis* in Baluran NP is domestic/feral and Not Evaluated by the IUCN Red List; C4 was set to 1 to reflect limited umbrella value for native biodiversity rather than global extinction risk.

In C5, "Priority species" denotes species currently prioritized in park management plans; it does not imply the focal-species concept in the theoretical sense.

Appendix S5 — Pairwise site-level co-occurrence (hypergeometric `cooccur`, R)

Table S5.1. Significant pairs only (positive, p_gt ≤ 0.05). No negative pairs were detected at α = 0.05.

Species 1	*Species 2*	n_i	n_j	*O (joint stations)*	*E (expected)*	*Δ (O–E)*	*p_lt*	*p_gt*
B. bubalis	*R. t. russa*	6	12	6	1.29	4.71	1.000	0.0000285
B. bubalis	*M. f. fascicularis*	6	11	5	1.18	3.82	1.000	0.000655
M. f. fascicularis	*R. t. russa*	11	12	7	2.36	4.64	1.000	0.000767
B. j. javanicus	*R. t. russa*	13	12	6	2.79	3.21	0.997	0.022
M. f. fascicularis	*M. m. muntjak*	11	11	5	2.16	2.84	0.996	0.029
R. t. russa	*S. scrofa*	12	7	4	1.50	2.50	0.997	0.032
B. j. javanicus	*S. scrofa*	13	7	4	1.62	2.38	0.995	0.043
M. m. muntjak	*R. t. russa*	11	12	5	2.36	2.64	0.992	0.045

Notes: *p_gt* = one-tailed probability that co-occurrence is higher than expected under independence (positive association); *p_lt* = one-tailed probability that co-occurrence is lower than expected under independence (negative association).

Table S5.2. For each candidate: C1 (% park-wide) = mean of O_{ij}/N across other mammals (diagonal excluded; N = 56), and Score 1–3 (tertiles; tie-friendly). Thresholds for this dataset: Q1 = 3.1994 %, Q2 = 4.9107 % (a value exactly equal to Q2 is assigned Score 2).

Species	*C1 (% park-wide)*	*Score (1–3; tertiles, tie-friendly)*	*#Stations detected (n_i)*	*Mean (%)*	*SD (%)*	*Range (%)*
R. t. russa	7.14	3	12	46.8	27.2	0–100
M. f. fascicularis	5.80	3	11	37.3	24.4	0–83
M. m. muntjak	4.92	3	11	29.1	15.6	0–45
B. j. javanicus	4.91	2	13	30.5	18.4	0–57
B. bubalis	3.57	2	6	17.9	18.6	0–50
S. scrofa	3.57	2	7	19.6	13	0–33
T. auratus	2.46	1	7	17.7	8.1	8–33
C. a. javanicus	2.01	1	6	14.8	10.4	0–33
P. p. melas	0.45	1	3	3.9	6.7	0–17

Notes: Co-occurrence analysis script (`cooccur`, R) (Griffith et al., 2016) available in the data repository: https://github.com/drajat313/baluran-umbrella-spp-data

Appendix S6 — Weight-sensitivity analysis of the seven-indicator framework.

Scope and input.

We evaluated the nine mammal candidates carried forward in the main text using the same seven indicators (C1–C7) and ordinal 1–3 scores described there (Table 1; Methods "Species evaluation framework"). The analysis used a species-by-indicator table containing scientific names and ordinal scores (1–3) for C1–C7, following Table 1 in the main text.

Weighting scenarios tested.

We evaluated ranking sensitivity under four weighting scenarios applied to the indicator weights: (i) baseline (all indicators multiplied by 1), (ii) one-at-a-time double-weight (the focal indicator multiplied by 2 while all others were multiplied by 1), (iii) leave-one-out (focal indicator multiplied by 0; others by 1), and (iv) random weights drawn from a symmetric Dirichlet(1,…,1) and normalised to sum to 1 (50,000 draws).

Ranking and tie handling.
In each run, species totals were computed as a weighted sum, $\boldsymbol{T_i} = \sum_k \boldsymbol{w_k C_{ik}}$, where T_i is the total score for species i, $C_{ik}$is species i's ordinal score (1–3) on indicator k, and $w_k$is the weight multiplier applied to indicator k. Only weight ratios affect ranks, so normalization is unnecessary for deterministic scenarios; Dirichlet weights are already normalized to sum to 1. Ties were handled by splitting credit evenly among tied species. For example, if two species tied for Top-1 in a given run, each received 0.5 credit. The same principle was applied when summarizing Top-2 frequencies.

Software and reproducibility.
Analyses were conducted in R 4.5.1 using dplyr, tibble, and tidyr, with writexl used only for optional Excel export. Random weight vectors were generated with set.seed(123) and Gamma normalization to sample from a Dirichlet distribution. The script provided in the Code listing computes weighted totals, identifies Top-1 and Top-2 ranks while accounting for ties, runs the sensitivity analyses, and summarizes frequencies and win counts.

Primary outputs.
The sensitivity analysis generated four complementary sets of results: baseline rankings under equal weights, rankings from the one-way double-weight scenarios, rankings from the leave-one-out scenarios, and Top-1/Top-2 frequencies from 50,000 random Dirichlet weight vectors. Together, these outputs were used to assess how consistently species remained highly ranked across alternative weighting schemes and therefore to evaluate the robustness of the umbrella-species ranking. Detailed results are reported in Appendices S6 and S7, and the full script is provided in the Code listing.

Limitations.
Results inherit the scope of the seven indicators and the camera-trap sample (medium–large terrestrial mammals); they do not generalize to other taxa without dedicated data. Equal-weight scoring was chosen for transparency; this appendix shows that key conclusions are not artifacts of a single weight choice.

Code availability (R).
A concise, annotated script implementing all steps is provided in the data repository:
https://github.com/drajat313/baluran-umbrella-spp-data

Appendix S7 — Robustness of rankings to random weights (Dirichlet simulations)

Table S7. Top-1 and Top-2 frequencies under 50,000 random weight vectors drawn from Dirichlet(1,…,1). Ties share credit for Top-1 and Top-2. Values are based on the nine candidate species and seven indicators used in the main analysis.

species	***top1_frequency_over_50000***	***top2_frequency_over_50000***
B. j. javanicus	0.18242	0.43916
P. p. melas	0.2664	0.50238
C. a. javanicus	0.08644	0.2033
M. m. muntjak	0.16314	0.30168
R. t. russa	0.14294	0.25866
S. scrofa	0.15774	0.28522
B. bubalis	0	0
M. f. fascicularis	0.00092	0.0096
T. auratus	0	0

Across neutral weight preferences, *P. p. melas* and *B. j. javanicus* most frequently ranked first and within the top two, indicating stable support for a two-species umbrella. Species with consistently high site-level co-occurrence, such as *R. t. russa* and *M. m. muntjak*, also featured prominently in Top-2 outcomes, reinforcing their role as complementary open-habitat indicators.

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